Section A – Answer any 2 out of 3 questions in section A

Question 1

a)

(i) What are the probabilities of meeting a peak demand of 148 kW, 105 kW and 97 kW respectively?

[5]

The availability table of generators:

Availability (kW)	*Probability (%)*
150	*94.11*
110	*0.95*
100	*1.92*
90	*2.91*
60	*0.02*
50	*0.03*
40	*0.06*

[4 marks]

The probabilities to meet peak demands are

94.11%, 95.06% and 96.98% respectively

[1 mark]

(ii) Assume that the coincidence factor of an infinite number of households is 0.444. The peak demand of each individual household is 5 kW and the reliability criterion requires that the probability of meeting peak demand is above 95%, how many households can be supplied by these generators?

[5]

To meet 95% reliability requirement, the maximum peak demand can be served is 110 kW

[2 Mark]

$$\left(0.444 + \frac{1-0.444}{\sqrt{n}}\right) n \times 5 = 110$$

$$\therefore n = 41.48$$

41 households

[3 marks]

b)

(i) Explain why faster frequency reserve (such as EFR) is more effective in limiting the frequency nadir

[2]

Model Answers

RoCoF has the higher value immediately after the disturbance. Hence, faster response helps to reduce the frequency deviation more effectively.

[2 mark]

(ii) Calculate how much EFR is needed to maintain the nadir constraint if the purchased frequency response (R) by system operator is 3000MW respectively 2500MW ?

[4]

1) $EFR = 1.8 - 0.025R - 0.5\sqrt{0.05^2R^2 + 3.48R}$

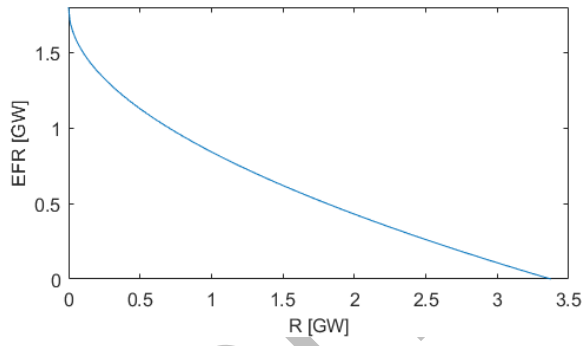
[1 mark]

2) R=3000MW, EFR =107.71MW,

R=2000MW, EFR =429.96MW,

[1 mark]

3)



[2 mark]

(iii) If the EFR price is 15 £/MW, determine the optimal amount of EFR and R the system operator should purchase to achieve the minimum cost?

[4]

$$\min_{EFR,R} 5R + 15EFR$$
$$s.t. EFR = 1.8 - 0.025R - 0.5\sqrt{0.05^2R^2 + 3.48R}$$

[1 mark]

$$\frac{\partial EFR}{\partial R} = -0.025 - 0.25(0.05^2R^2 + 3.48R)^{-0.5}(0.05^2 2R + 3.48) = -\frac{5}{15}$$

[1 mark]

$$R = 2.30\ GW$$
$$EFR = 0.327\ GW$$

[2 mark]

Model Answers

Question 2

a) What are the steps to follow when converting to a per-unit system in the analysis of power systems?

For the entire system, choose arbitrarily a power basis
For each voltage level, choose a voltage basis (nominal voltage is recommended)
Then, for each voltage level, calculate the current and impendent basis

[1 mark each, 3]

b)

(i) Calculate the per unit values of the quantities (power and impedance) in the figure, with proper selected bases.

Choose: $S_B = 50MW, V_{B11} = 11kV,\ V_{B132} = 132kV, V_{B33} = 33kV$

[1 mark]

$P_s = 1\ p.u.\ \ Q_r = 0.8\ p.u.$

[1 mark]

$X_1 = j0.10$

$X_2 = \dfrac{j100}{V_{B132}^2/S_B}$

$X_3 = j0.12\dfrac{50}{60}$

[1 mark]

(ii) Calculate Vs and Qs in kV and MVar respectively.

$|V_r| = \left|\frac{S_r}{I_r}\right| = 0.97$, *(set as a angle reference*, $V_r = 0.97\angle 0°$*)*

[1 mark]

$I_r = \left(\frac{S_r}{V_r}\right)^* = 1.32\angle -38.66°$

[1 mark]

$V_s = V_r + I_r X = 0.97\angle 0° + 1.32\angle -38.66°\ j0.487 = 1.46\angle 20.10°$

[1 mark]

$V_s = 1.46V_{B11} = 16.06kV$

$S_s = V_s I_s^* = S_B * 1.46\angle 20.10° * 1.32\angle 38.66° = 50MW + j82.5MVar$

[1 mark]

Same active power since R=0, reduced reactive power due to inductive line.

[1 mark]

c) Write down the general power flow equations of a two-bus system (seen from the sending end, neglecting the shunt admittance). Derive the sensitivities of the active and reactive power with respect to the angle difference and the sending end voltage magnitude (four elements). What conclusions can be made if the line resistance is zero, regarding the PQ-Vθ coupling? Further, assume that the voltage magnitudes

Model Answers

at both ends are 1 pu and the voltage angle difference is small. Derive the DC approximation of the power flow equations.

power flow

[2 mark]

Jacobian matrix

[2 mark]

PQ decoupling

[1 mark]

DC power flow

[2 mark]

DC power flow, matrix form, $P = B\theta$

[2 mark]

Question 3

a)

$$i_1 = j\frac{B}{2}V_1 + \frac{1}{Z_1}(V_1 - V_2) + \frac{1}{Z_2}(V_1 - V_3) = \left(j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_2}\right)V_1 - \frac{1}{Z_1}V_2 - \frac{1}{Z_2}V_3$$

$$i_2 = j\frac{B}{2}V_2 + \frac{1}{Z_1}(V_2 - V_1) + \frac{1}{Z_3}(V_2 - V_3) = -\frac{1}{Z_1}V_1 + \left(j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_3}\right)V_2 - \frac{1}{Z_3}V_3$$

$$i_3 = \frac{1}{Z_2}(V_3 - V_1) + \frac{1}{Z_3}(V_3 - V_2) = -\frac{1}{Z_2}V_1 - \frac{1}{Z_3}V_2 + \left(\frac{1}{Z_2} + \frac{1}{Z_3}\right)V_3$$

[1 mark each, 3]

$$\therefore Y = \begin{pmatrix} j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_2} & -\frac{1}{Z_1} & -\frac{1}{Z_2} \\ -\frac{1}{Z_1} & j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_3} & -\frac{1}{Z_3} \\ -\frac{1}{Z_2} & -\frac{1}{Z_3} & \frac{1}{Z_2} + \frac{1}{Z_3} \end{pmatrix}$$

[2 marks]

b)

[12]

$$V_1 = 1\angle 0° \qquad V_2^0 = 1.0\angle 0° \qquad V_3^0 = 1.04\angle 0°$$

[1 mark]

$$Y = \begin{pmatrix} -j8 & j4 & j4 \\ j4 & -j9 & j5 \\ j4 & j5 & -j9 \end{pmatrix}$$

[1 mark]

Model Answers

Iteration 1:

$$i_2 = \frac{S_2^*}{V_2^*} = Y_{21}V_1 + Y_{22}V_2 + Y_{23}V_3$$

$$V_2^1 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{0*}} - Y_{21}V_1 - Y_{23}V_3^0\right) = \frac{1}{-j9}\left(\frac{-0.97 + 1j}{1} - j4 - j5 \times (1.04)\right)$$
$$= 0.9174\angle -6.7463°$$

[2 marks]

$$S_3^* = V_3^{0*} i_3 = V_3^{0*}(Y_{31}V_1 + Y_{33}V_3^0 + Y_{32}V_2^1)$$
$$= 1.04 \times (j4 - j9 \times 1.04 + j5 \times (0.9174\angle -6.7463°))$$
$$= 0.5607 - j0.8370$$

$$Q_3 = j0.8370$$

[2 marks]

$$V_3^1 = \frac{1}{Y_{33}}\left(\frac{S_3^*}{V_3^{0*}} - Y_{31}V_1 - Y_{32}V_2^1\right)$$
$$= \frac{1}{-j9}\left(\frac{1.02 - j0.8370}{1.04} - j4 - j5(0.9174\angle -6.7463°)\right)$$
$$= 1.0412\ \angle 2.7013°$$

[2 marks]

Iteration 2:

$$V_2^2 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{1*}} - Y_{21}V_1 - Y_{23}V_3^1\right)$$
$$= \frac{1}{-j9}\left(\frac{-0.97 + 1j}{0.9174\angle 6.7463°} - j4 - j5 \times (1.0412\ \angle 2.7013°)\right)$$
$$= 0.8913\ \angle -4.8386°$$

[1 mark]

$$S_3^* = V_3^{1*} i_3 = V_3^{1*}(Y_{31}V_1 + Y_{33}V_3^1 + Y_{32}V_2^2)$$
$$= 1.04\angle$$
$$-2.7013° \times (j4 - j9 \times 1.04\angle 2.7013° + j5 \times (0.8913\ \angle -4.8386°))$$
$$= 0.8040 - j0.9843$$

$$Q_3 = j0.9843$$

[1 mark]

$$V_3^2 = \frac{1}{Y_{33}}\left(\frac{S_3^*}{V_3^{1*}} - Y_{31}V_1 - Y_{32}V_2^2\right)$$
$$= \frac{1}{-j9}\left(\frac{1.02 - j0.9843}{1.04\angle -2.7013°} - j4 - j5(0.8913\ \angle -4.8386°)\right)$$
$$= 1.0403\ \angle 3.9712°$$

[2 marks]

Model Answers

c) The performance of the Loads that are sensitive to voltage magnitude can be compromised.

[1 mark]
Any means to effectively increase V3 or decrease Q2
[2 mark]

Model Answers

Question 4

a) The four steps are:

1. Obtain the pre-fault bus voltages and line currents from power flow analysis.
2. Replace the generators and motors by their sub-transient reactance connected to ground.
3. Excite the passive Thevenin equivalent (all voltage sources shorted) at the fault bus by the negative of the pre-fault voltage in series with fault impedance and compute change in voltage and current due to fault.
4. Obtain the bus voltage and line current during fault by adding (superposition) of the results of steps 1 (pre-fault) and 3 (sub-transient).

[1 mark for each step – 4 marks]

b)

- Short circuit level (in MVA) at a particular bus is the apparent power (MVA) due to a zero-impedance fault current I_f (in kA) under nominal system voltage V_{TH} (kV)

  $$\text{SCL (in MVA)} = \sqrt{3} V_{TH} I_f$$

  $$\text{SCL (p.u.)} = \frac{\text{SCL (MVA)}}{S_{\text{base}}} = \frac{I_f}{I_{\text{base}}} = I_f \text{ (p.u.)} = \frac{V_{TH}\text{(p.u.)}}{Z_{TH}\text{(p.u.)}} \approx \frac{1.0}{Z_{TH}\text{(p.u.)}}$$

- System strength at a particular bus is the ability to regulate the bus voltage as active and/or reactive power exchange through that bus vary.
- For a given active/reactive power exchange voltage regulation is poor when the equivalent impedance at that bus Z_{TH} is relatively high i.e., the short circuit level is low.
- Hence, if the short circuit level is large (small), the system at that bus is "strong" ("weak") i.e., the system strength at a particular bus is dictated by the short circuit level.

[1 mark for each point – 4 marks]

c)

- Following a fault, a synchronous generator offers a time varying reactance with the reactance being initial low and gradually increasing to its steady state value with passage of time.
- Due to constant flux linkage principle, the excess flux due to the fault current can't penetrate the field and damper winding immediately after the fault and is forced through high reluctance paths that do not link the field or damper.
- Thus, the stator inductance which is inversely proportional to reluctance is initially low which is known as sub-transient reactance.
- With passage of time, as the excess lux penetrates the damper and field which are lower reluctance paths, stator inductance gradually increases to its transient and then, steady state value.
- As fault calculation relates to the time immediately after the fault, sub-transient reactance of generator is used instead of steady-state reactance which would give lower than actual fault current.

[1 mark for each point – 5 marks]

Model Answers

d)

i)

System base $S_b = 50$ MVA

Generator sub-transient reactance in p.u. on system base

$$X''_{G1} = X''_{G1} \times \frac{S_b}{S_{G1}} = 0.1 \times \frac{50}{45} = 0.11 \text{ p.u.}$$

$$X''_{G2} = X''_{G2} \times \frac{S_b}{S_{G2}} = 0.15 \times \frac{50}{50} = 0.15 \text{ p.u.}$$

$$X''_{G3} = X''_{G3} \times \frac{S_b}{S_{G3}} = 0.1 \times \frac{50}{40} = 0.125 \text{ p.u.}$$

[1 mark]

Reactance of the reactors in p.u. on system base

$$X_{12} = X_{12} \times \frac{S_b}{S_{12}} = 0.22 \text{ p.u.}$$

$$X_{23} = X_{23} \times \frac{S_b}{S_{23}} = 0.375 \text{ p.u.}$$

[1 mark]

Equivalent reactance (X_{eq}) at the fault location

$$\frac{1}{X_{eq}} = \frac{1}{X''_{G1} + X_{12}} + \frac{1}{X''_{G2}} + \frac{1}{X''_{G3} + X_{23}}$$

$$X_{eq} = 0.0857 \text{ p.u.}$$

[2 marks]

Short circuit level

$$SCL = \frac{S_b}{X_{eq}} = 583.3 \text{ MVA}$$

[1 mark]

ii)

Base current

$$I_b = \frac{S_b}{\sqrt{3}V_b} = 1.44 \text{ kA}$$

[1 mark]

Fault current (in p.u.)

$$I_f(\text{p.u.}) = \frac{1}{X_{eq}} = 11.67 \text{ p.u.}$$

$$I_f(\text{in kA}) = I_f(\text{p.u.}) \times I_b = 16.84 \text{ kA}$$

[1 mark]

Model Answers

Question 5

a)

i) LG fault

Fault current and voltage in phase domain:

$I_b = I_c = 0, \quad V_a = I_a Z_f$

[1 mark]

Fault current and voltage in sequence domain:

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} I_a \\ 0 \\ 0 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} I_a \\ I_a \\ I_a \end{bmatrix}$$

[1 mark]

$$I_{a0} = I_{a1} = I_{a2} = \frac{1}{3} I_a$$

[1 mark]

$V_a = V_{a0} + V_{a1} + V_{a2} = I_a Z_f = 3 I_{a0} Z_f$

[1 mark]

As the current through the positive, negative and zero sequence networks is the same as the sum of the voltages across them is $3I_{a0}Z_f$, the sequence networks are connected in series with 3 times the fault impedance.

[1 mark]

ii) LL fault

Fault condition in phase domain

$I_a = 0, I_c = -I_b, V_b - V_c = Z_f I_b$

[1 mark]

Fault current and voltage in sequence domain:

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = [\mathbf{A}^{-1}]\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} 0 \\ I_b \\ -I_b \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 0 \\ (\alpha - \alpha^2) I_b \\ -(\alpha - \alpha^2) I_b \end{bmatrix}$$

[1 mark]

$$I_{a1} = -I_{a2} = \frac{1}{3}(\alpha - \alpha^2) I_b = j\frac{I_b}{\sqrt{3}}, \quad I_{a0} = 0$$

Model Answers

[1 mark]

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} == \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_b - Z_f I_b \end{bmatrix}$$

$$V_{a1} - V_{a2} = \frac{(\alpha - \alpha^2)}{3} Z_f I_b = Z_f I_{a1}$$

[1 mark]

Zero sequence network is not involved as $I_{a0} = 0$. Currents through the positive and negative sequence network are equal and opposite and their voltage difference is equal to the voltage drop across the fault impedance. Hence, the positive and negative sequence networks are connected in parallel across the fault impedance.

[1 mark]

b)

i)

Equivalent reactance of positive, negative and zero sequence networks at fault point are

$X_1 = X''_{G1} + X_{T1} = 0.15$ p.u.

$X_2 = X''_{G2} + X_{T2} = 0.15$ p.u.

[1 mark]

Zero sequence current won't flow on the generator side of the delta winding of transformer

$X_0 = X_{T0} = 0.1$ p.u

[1 mark]

LG fault current

$$I_{LG} = \frac{3}{X_1 + X_2 + X_0} = 7.5 \text{ p.u.}$$

[1 mark]

ii)

With LG fault on the generator terminal, the transformer won't play a part. Hence, the equivalent reactance of the sequence networks at fault point are

$X_1 = X''_{G1} = 0.05$ p.u.

$X_2 = X''_{G2} = 0.05$ p.u.

$X_0 = X''_{G0} = 0.01$ p.u

[1 mark]

Model Answers

LG fault current

$$I_{LG} = \frac{3}{X_1 + X_2 + X_0} = 27.27 \text{ p.u.}$$

[1 mark]

c)

i)

When both sides of the transformer are star connected with solidly grounded neutral, zero sequence current can flow through both generator and transformer. Hence, the equivalent reactance of the sequence networks at the fault point are:

$X_1 = X''_{G1} + X_{T1} = 0.15$ p.u.

$X_2 = X''_{G2} + X_{T2} = 0.15$ p.u.

$X_0 = X''_{G0} + X_{T0} = 0.11$ p.u

[1 mark]

LG fault current

$$I_{LG} = \frac{3}{X_1 + X_2 + X_0} = 7.32 \text{ p.u.}$$

[1 mark]

ii)

For LG fault on the generator terminal, the transformer won't play a part. So the LG fault current would be 27.27 p.u. – same as part b(ii) – even after the 11 kV side of the transformer is change to star connection with solidly grounded neutral.

[1 mark]

iii)

- Change of connection type of transformer winding affects the equivalent impedance of the zero-sequence network but not positive and negative sequence networks.
- The zero-sequence network does not play a part for line to line (LL) faults. Hence, LL fault current won't be affected by change of transformer connection type.

[1 mark for each point – 2 marks]

Model Answers

Question 6

a)
For power angle less than 90 degrees in steady state:

- An increase in mechanical input leads to acceleration of the rotor resulting in increase in power angle and hence electrical power output.
- Thus, an increase in mechanical power input implies corresponding increase of electrical power output which helps maintain stability.

- However, for power angle more than 90 degrees in steady state:
  - An increase in mechanical input results in increase in power angle as before, but on this occasion increase in power angle leads to reduction in electrical power output because of the negative slope of power angle curve.
  - This is a runway situation with increase in mechanical input causing reduction in electrical power output and hence, the situation is unstable.

[1 mark for each point – 4 marks]

b)

Incremental changes in electrical power output:

$$\Delta P_e = \left(\frac{\partial P_e}{\partial \delta}\right)_0 \Delta\delta$$

Considering incremental changes from swing equation

$$\frac{2H}{\omega_S}\frac{d^2\Delta\delta}{dt^2} = -\Delta P_e$$

[1 mark]

Combining the above the characteristic equation is

$$\frac{2H}{\omega_S}\frac{d^2\Delta\delta}{dt^2} + \left(\frac{\partial P_e}{\partial \delta}\right)_0 \Delta\delta = 0$$
$$\Rightarrow \frac{2H}{\omega_S}s^2 + \left(\frac{\partial P_e}{\partial \delta}\right)_0 = 0$$

[1 mark]

$$s = \pm\left[-\frac{\left(\frac{\partial P_e}{\partial \delta}\right)_0 \omega_S}{2H}\right]^{\frac{1}{2}}$$

[1 mark]

For $\delta < 90°$, $\left(\frac{\partial P_e}{\partial \delta}\right)_0 > 0$ i.e., the poles are on the imaginary axis resulting in a critically damped stable system.

For $\delta > 90°$, $\left(\frac{\partial P_e}{\partial \delta}\right)_0 < 0$ i.e., one real pole is on the right half resulting in an unstable system.
[1 mark]

Model Answers

c)

i)

$P_{max} = \frac{1}{X} = 3.33$ p.u.

$P = 2$ p.u.

$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max}}\right) = 36.87°$ (0.64 rad)

$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 71.77°$ (1.25 rad)

[1 mark for each step – 4 marks]

ii)

$\frac{P'_{max}}{P_{max}} = \frac{X}{X(1-k)} = 1.5$

$k = 0.33$ or 33%

[2 marks]

iii)

$P_{max1} = \frac{1}{X(1-k)} = 5$ p.u.

[1 mark]

$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max1}}\right) = 23.58°$ (0.41 rad)

$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 89.38°$ (1.56 rad)

[1 mark]

iv)

Under the condition specified in the problem, the critical clearing angle (δ_{cr}) depends only on the initial power angle (δ_0) which depend on $\frac{P}{P_{max}}$

For part (i)

$\frac{P}{P_{max}} = \frac{2}{3.33} = 0.6$

[1 mark]

If P' is the power transfer possible with the same critical clearing angle (δ_{cr}) as in part (i)

$\frac{P'}{P_{max1}} = 0.6 \rightarrow P' = 0.6 \times P_{max1} = 3.0$ p.u.

Model Answers

[1 mark]

d)

The statement signify that the generator can sustain its rated power output for 6 seconds based solely on the kinetic energy stored in the rotor at synchronous speed.

[2 marks]

Model Answers